



Protein research in millets: current status and way forward

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Abstract

Main conclusion Millets' protein studies are lagging behind those of major cereals. Current status and future insights into the investigation of millet proteins are discussed.

Abstract Millets are important small-seeded cereals majorly grown and consumed by people in Asia and Africa and are considered crops of future food security. Although millets possess excellent climate resilience and nutrient supplementation properties, their research advancements have been lagging behind major cereals. Although considerable genomic resources have been developed in recent years, research on millet proteins and proteomes is currently limited, highlighting a need for further investigation in this area. This review provides the current status of protein research in millets and provides insights to understand protein responses for climate resilience and nutrient supplementation in millets. The reference proteome data is available for sorghum, foxtail millet, and proso millet to date; other millets, such as pearl millet, finger millet, barnyard millet, kodo millet, teff, and browntop millet, do not have any reference proteome data. Many studies were reported on stress-responsive protein identification in foxtail millet, with most studies on the identification of proteins under drought-stress conditions. Pearl millet has a few reports on protein identification under drought and saline stress. Finger millet is the only other millet to have a report on stress-responsive (drought) protein identification in the leaf. For protein localization studies, foxtail millet has a few reports. Sorghum has the highest number of 40 experimentally proven crystal structures, and other millets have fewer or no experimentally proven structures. Further proteomics studies will help dissect the specific proteins involved in climate resilience and nutrient supplementation and aid in breeding better crops to conserve food security.

Keywords Millet proteomics · Stress-responsive proteins · Protein localization · Protein structure · Proteome data

Introduction

Millets have been the important and stable cereal crops of the semi-arid tropics of Asia and Africa for several thousand years. Millets are a group of small-seeded cereal crops, mostly grown on unimproved soils without proper irrigation and fertilization (Goron and Raizada 2015). The small or minor millets include foxtail millet (*Setaria italica* (L.) P. Beauv.), finger millet (*Eleusine coracana* (L.) Gaertn.), proso millet (*Panicum miliaceum* L.), barnyard millet (*Echinochloa crus-galli* (L.) P. Beauv. and *Echinochloa colona* (L.) Link), little millet (*Panicum sumatrense* Roth. ex. Roem. & Schult.), kodo millet (*Paspalum scrobiculatum* L.), teff (*Eragrostis tef* (Zucc.) Trotter], fonio (*Digitaria exilis* Stapf and D. iburua Stapf.), job's tears (*Coix lacryma-jobi* L.), guinea millet (*Brachiaria deflexa* (Schumacher.) C.E.Hubb. ex Robyns, = *Urochloa deflexa* (Schumacher.) H.Scholz), and browntop millet (*Brachiaria ramosa*

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(L.) Stapf. = *Urochloa ramosa* (L.) T.Q. Nguyen). While sorghum (*Sorghum bicolor* L. Moench) and pearl millet (*Cenchrus americanus*) are called major millets (Vetriventhan et al. 2020). Although millets have been cultivated and consumed since ancient times, the original base was lost in the twentieth century due to the green revolution-induced proliferation of other major stable crops like rice and wheat (Antony Ceasar and Maharajan 2022). However, the millets have been given much importance and revitalized in recent years due to their excellent climate resilience and nutrient supplementation features. As a result, the millets have been proposed to help strengthen food and nutrient security, especially in the face of climate change (Saxena et al. 2018). Even millets are projected to help strengthen several key sustainable developmental goals (SDGs) of the United

Nations (Antony Ceasar and Maharajan 2022; Wang et al. 2018). Millets are a source of excellent mineral nutrients, vitamins, and important secondary metabolites. They are high in fiber and gluten-free. The mineral-nutrient composition of millets is superior to that of major cereals like rice and wheat (Nithiyanantham et al. 2019). It is also important to underscore that the calcium content of finger millet is tenfold higher than that of other cereals, which helps protect bones and teeth (Maharajan et al. 2021).

Even though millets possess excellent climate resilience features on the field and provide enduring nutritional benefits to the human diet, their research advancements have been lagging behind major cereals (Ajeesh Krishna et al. 2022). The omics (genomic, transcriptomic, metabolomic, and proteomic) studies of millets are lagging behind those

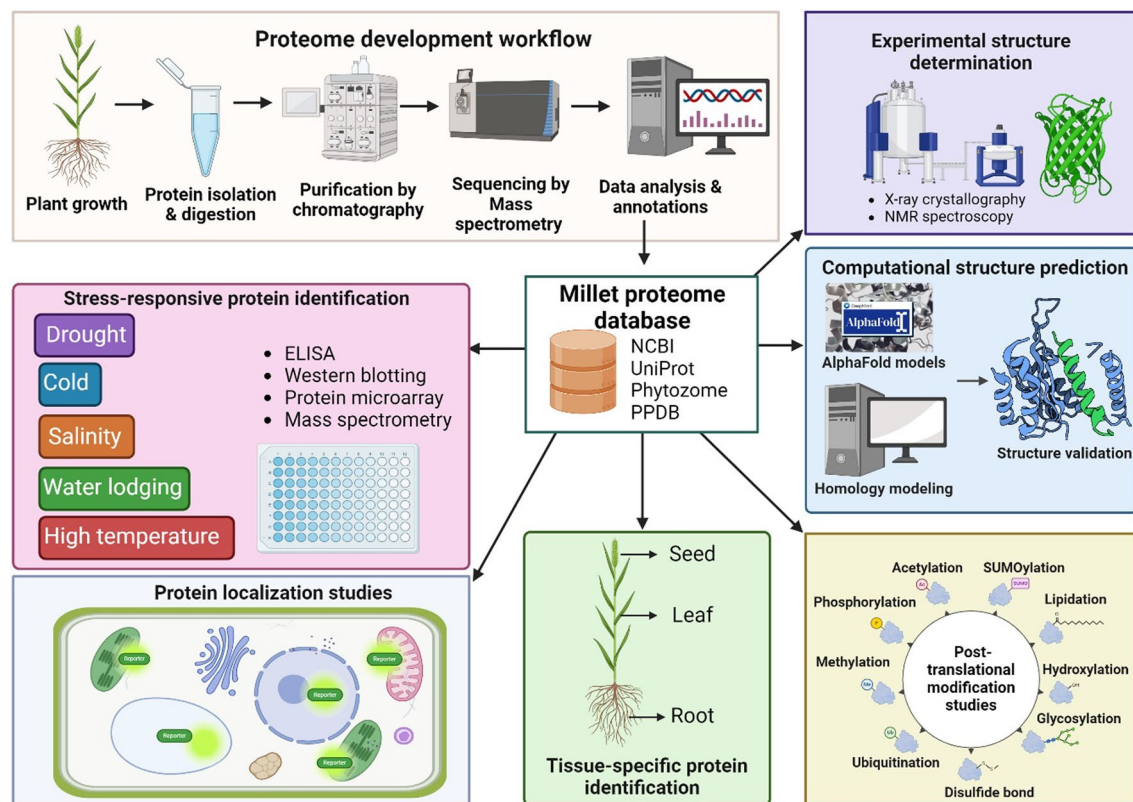


Fig. 1 Proteome data generation and its applications. The work flow starts from isolating proteins from specific tissues of the plants grown under specific conditions. This includes targeted extraction methods and to enrich for desired protein populations. The proteins are then digested by specific protease. The complex protein mixtures are separated using techniques like gel electrophoresis, or liquid chromatography. Following separation, mass spectrometry is used to identify the individual peptides within each fraction. Finally, bioinformatics tools are applied to match these identified peptides back to the predicted proteins encoded in the plant's genome data and to generate the proteome data. The proteome data are mostly deposited and freely available from NCBI (<https://www.ncbi.nlm.nih.gov/>), UniProt (<https://www.uniprot.org/>), Phytozome (<https://phytozome-next.jgi.doe.gov/>)

and PPDB (<http://ppdb.tc.cornell.edu/>), web-enabled search websites. All other downstream studies with millet proteomics are possible with proteome data. The stress-responsive proteins can be identified under various stresses such as drought, cold, salinity, and water lodging. Proteins expressed in different organs like the root, shoot, leaf, and developing spike can be identified under different stress conditions. Localization of proteins inside the cell can be performed with the fusion of reported proteins such as GUS and GFP and by visualization. Protein post-translational modification can also be studied. Protein structure can be identified by experimental methods such as X-ray crystallography. Models of protein structures can quickly be obtained by homology modeling and using the AI-based AlphaFold tool

of other major cereals. Proteomics is the large-scale study of proteins, focusing on their structures, functions, and interactions. It involves analyzing the sequence, structure, function, expression, modifications, and interactions of proteins to understand cellular processes (Fig. 1). A proteome is the group of proteins that an organism, cell, tissue, or organelle expresses at a specific time under a specific condition. UniProt tackles the exploding number of proteomes by creating a curated set of reference proteomes. Reference proteomes, as defined by UniProt (https://www.uniprot.org/help/reference_proteome), serve as key landmarks in proteome space, aiding navigation through the growing number of sequenced genomes. Reference proteomes are selected based on criteria ensuring broad coverage of the tree of life, representing a diverse cross-section of taxonomic diversity. They include well-studied model organisms and important species for biomedical and biotechnological research, sometimes featuring multiple proteomes for specific ecotypes or strains. It serves as a standardized dataset for studying the biology of an organism. It provides a benchmark for identifying and annotating proteins in related species or strains, facilitating comparative proteomics. Many omics resources have been developed only in recent years for millets (Jadhav et al. 2024; Maharajan et al. 2024a, b). Foxtail millet was the first millet to have its complete genome sequence released in 2012 with complete annotations (Bennetzen et al. 2012; Zhang et al. 2012). Finger millet is the only other millet to have a completely annotated genome sequence released in 2023 (Devos et al. 2023). A high-quality and chromosome-level reference genome assembly was recently released for pearl millet (Ramu et al. 2023). Proso millet and barnyard millet, have only draft genome sequences without complete annotations to date. Similarly, other genetic and biotechnology resources, such as the development of transformation protocols, transgenic plants, and genome editing of millets, are all lagging behind other major cereals (Ceasar 2021). Millets have not yet been explored for protein research, and only a limited report is available on proteomic studies in millets to date. It is important to develop proteomic tools and investigate millet proteins along with other omics tools that would help to dissect the mechanisms of climate resilience and nutrient enrichment in millets.

Nearly every biological function is executed and regulated by proteins in close proximity (Mergner and Kuster 2022). Proteins play a critical role in mediating the stress responses of plants. Proteomics tools enabled scientists to analyze protein profiles in stressed plants, revealing the potential connection between proteins and plant stress resistance (Rakkammal et al. 2024) (Fig. 1). Investigating the molecular mechanism of stress tolerance can help progress the crop improvement program (Subramani et al. 2024). Although the significance of RNA sequencing (RNAseq) as a transcriptomic technology, the RNAseq findings alone

may not provide a comprehensive insight into the proteins expressed from the transcriptome and specifically enhanced within a particular tissue during specific stress conditions. This limitation arises from the numerous instances of alternative splicing, which can significantly alter the proteome landscape under such conditions (Zhong et al. 2024). Millets, despite their well-known climate resilience properties, remain under-explored in terms of protein research for stress conditions. Millets have not received the same level of research focus as major cereal crops like wheat and rice. So, millets have fewer genetic tools and databases available for millet research, making proteomic studies more challenging. Modern proteomic techniques can easily reveal the molecular underpinnings of stress responses and analyze the functionality of translated genomic regions. Identifying and verifying stress-responsive proteins is a viable technique for breeding better crops in the future (Liu et al. 2015). Modern proteomic techniques will be helpful for unlocking the protein responses of stress resilience and nutrient enrichment in millets. Especially by analyzing the proteins that increase or decrease under stress conditions (like drought or high temperatures), researchers can identify the specific proteins responsible for the crop's resilience (Fig. 1). This knowledge can then be used to develop targeted breeding programs that enhance these stress tolerance mechanisms in millets and other crops. Proteomic tools can help pinpoint the specific proteins involved in nutrient storage in millet seeds. Investigating these proteins allows researchers to create biofortified millet varieties with even higher levels of essential minerals and vitamins.

In this article, the current status of protein research, including protein sequences, proteome data, protein identification in various physiological conditions, protein localization, and structure determination studies in millets, has been reviewed. The future direction and insights for millet protein research are discussed. Identifying and utilizing millet proteins would aid in identifying the primary contributors to climate resilience and nutrient enrichment in millets.

Protein research in millets

There is currently a lack of extensive research on millet proteins and proteomes, indicating a requirement for additional exploration in this field. The availability of reference proteome data is a vital prerequisite for designing many proteomic studies of a crop. More than one reference proteome is available for several model plants, like rice and maize (Table 1). Rice has three reference proteome data sets, and maize has two reference proteome data sets to date (<https://www.uniprot.org/>). Apart from reference proteome, organelle proteome identified through 2D gel electrophoresis is also available for rice (Mikami et al. 2002; Komatsu et al. 2004; Ambastha et al. 2020). Sorghum has one reference

Table 1 List of reference proteomes, UniProt protein sequences entries, experimentally proven crystal structures and AlphaFold models available for millets and major cereals

Name of the millet	Reference proteome	UniProt entries	Crystal structures (PDB)	AlphaFold models (PDB)
<i>Oryza sativa</i> (Rice) [#]	03	1,49,133	243	44,514
<i>Triticum aestivum</i> (Wheat)	02	1,52,266	106	372
<i>Zea mays</i> (Maize)	01	2,07,453	247	39,318
<i>Sorghum bicolor</i> (Sorghum)	01	82,121	40	179
<i>Cenchrus americanus</i> (Pearl millet)	0	541	07	03
<i>Setaria italica</i> (Foxtail millet)	02	70,501	4	03
<i>Panicum miliaceum</i> (proso millet)	1	55,825	01	01
<i>Eleusine coracana</i> (Finger millet)	0	509	3	02
<i>Echinochloa crus-galli</i> (Barnyard millet)	0	146	01	03
<i>Paspalum scrobiculatum</i> (Kodo millet)	0	0	0	0
<i>Panicum sumatrense</i> (Little millet)	0	88	01	0
<i>Eragrostis tef</i> (Teff)	0	210	0	0
<i>Brachiaria ramosa</i> (Browntop millet)	0	9	0	0

Details are obtained from the UniProt and PDB websites by typing respective species names

As per data accessed from <https://www.uniprot.org/> and <https://www.rcsb.org/> at 11–06–2024. The specific species name (*Oryza sativa* [both subspecies *japonica* and *indica*], *Triticum aestivum*, *Zea mays*, *Sorghum bicolor*, *Cenchrus americanus*, *Setaria italica*, *Panicum miliaceum*, *Eleusine coracana*, *Echinochloa crus-galli*, *Paspalum scrobiculatum*, *Panicum sumatrense*, *Eragrostis tef* or *Brachiaria ramosa*) was used as search key word at <https://www.uniprot.org/> and <https://www.rcsb.org/>

[#]Rice includes both subspecies *japonica* and *indica*

proteome data set to date. This reference proteome data has 41,380 proteins for strain cv. BTx623 (Mergner and Kuster 2022). For foxtail millet, a reference proteome data is available with 39,467 protein entries for strain cv. Yugu1, which was used for genome assembly (Bennetzen et al. 2012). Similarly, a reference proteome is available for the proso millet with 55,825 protein entries (Zou et al. 2019). Other millets, such as pearl millet, finger millet, barnyard millet, kodo millet, teff, and browntop millet, do not have any proteome data to date. The production of proteome data is a complex procedure that encompasses various advanced techniques such as biochemistry, mass spectrometry, and bioinformatics (Fig. 1). The procedure involves extracting proteins from the targeted tissues of the plants grown under specified conditions, such as leaf tissues from drought stress. Subsequently, specialized proteases break down these proteins into peptides. Mass spectrometry is employed to determine the mass-to-charge ratio (m/z) of each individual peptide. Finally, bioinformatics tools are utilized to compare the mass spectrometry data with extensive protein databases in order to ascertain the precise proteins that are present in a particular tissue (Rödiger et al. 2014; Jorin Novo 2021) (Valledor et al. 2014). By creating and analyzing proteome data, researchers can identify the specific proteins that are essential for stress tolerance or nutrient accumulation in plants, including millets. The proteome is also extremely dynamic in both space and time, depending on the conditions and stage of development. Apart from the individual

variations, several proteins are also involved in a wide range of protein–protein interactions (Mergner and Kuster 2022). Studying the proteome of millet will have great significance as millets, with their well-known resilience to harsh environments, likely exhibit unique protein shifts in response to stress conditions. Identification and analysis of key proteins under such conditions could contribute to developing stress tolerance in both millet and non-millet crops.

Although the reference proteome data sets available for millets are limited, a sizable number of protein sequences are available for several minor millets. Sorghum, which also has proteome-scale protein data, has 82,121 entries for protein. The foxtail millet has 70,501 entries among the minor millets. The little millet, with 35,076 entries, stands second among the minor millets for the number of protein sequence entries (<https://www.uniprot.org/>). Other millets have a few hundred or only a smaller number of protein entries at UniProt. Pearl millet has no reference proteome data to date and has only 541 entries in UniProt. Finger millet also has only 509 entries. Barnyard millet and teff have only a few over a hundred entries for proteins. Particularly, the kodo millet has zero entries for protein sequences. As discussed above, most of these minor millets have high climate resilience and nutrient supplementation features. Genes serve as the repository of information for protein synthesis, while proteins carry out numerous functions as the primary agents within the cell. Since many proteins undergo dynamic changes after translation and even trafficking, the availability of specific

proteins in specific tissues and organelles needs to be confirmed (Fig. 1). This will not correlate with genomic and transcriptomic data. So, studying millet proteins plays a vital role in dissecting the specific proteins and characterizing their role in climate resilience and nutrient supplementation functions. The reference proteome data of selective millets will help mine key proteins involved in such functions. More reference proteome data needs to be developed to carry out further characterization and identification of proteins in other millets for key traits.

Tissue-specific protein identification

The identification of tissue-specific proteins is crucial in proteomics research, which aims to uncover complex biological processes in the many tissues of an organism. The growth and development of plants depend on the tissue-specific production of proteins. Therefore, studies of tissue-specific proteins contribute to our knowledge of plant responses to abiotic stresses. In plants, each tissue has a unique and key function in growth and reproduction. For instance, photosynthesis is the primary function of the highly specialized tissues in leaves. Numerous studies have demonstrated that the rates and activities of photosynthesis are closely linked to the age and growth of the leaf (Kitajima et al. 1997; Wilson et al. 2001; Ebeed and El-Helely 2021). This is also the case with the accumulation of proteins connected to photosynthesis, such as ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCO) (Urban et al. 2008). Similar to this, root tissues are able to detect signals from the soil, such as water, nutrients, and gravity. Additionally, they can release a variety of substances that guard the root apex and control root growth (Feldman 1984) (Aiken and Smucker 1996). A better understanding of the mechanisms determining root length and branching is expected to improve crop production and yields (Lynch 2007). Abiotic stress initially affects the underground or over-ground part of a plant and is sensed by the roots or leaves that finally trigger cellular signal transduction pathways leading to molecular and metabolic changes. Hence, for a better understanding of how plants respond and adapt to abiotic stresses, it is important to focus on root and leaf tissue protein identification. So, a better understanding of the mechanisms determining root length and branching is expected to improve crop production/yields (Feldman 1984).

Tissue-specific proteins have been identified in some plants in response to stress conditions. Changes in tissue-specific protein levels were analyzed in the leaf, hypocotyl, and root tissues of drought-stressed soybean seedlings (Mohammadi et al. 2012). Proteins were also identified in the leaf, hypocotyl, and root tissues of soybean, in response to flooding and drought stress (Wang et al. 2017), and under flooding stress (Khatoon et al. 2012). Under flooding and

drought, the activities of alcohol dehydrogenase and delta-1-pyrroline-5-carboxylate synthase enzymes went up in soybean seedlings (Wang et al. 2017). Floral-specific proteins of an ornamental orchid (*Cymbidium goeringii*) was analyzed to find the floral development-associated proteins (Chen et al. 2021). Extracting and analyzing proteins from particular tissues requires a number of standardized steps (Fig. 1). A number of techniques for extracting and solubilizing proteins have been tested to provide repeatable, high-quality proteome maps of different plant tissues, despite the significant challenges involved in identifying tissue-specific proteins due to factors such as the complex nature of tissues, dynamic protein expression patterns, low abundance of certain proteins, and constraints of experimental methods (Jorrín et al. 2007). For example, RuBisCO, which is an abundant protein in the leaf tissues, was eliminated from other leaf proteins during extraction, a number of enrichment methods have been used due to presence of high levels of low-abundance proteins in root tissues, and phenol-based method gives reproducible, high-quality proteome maps for seed proteins of soybean compared to other available methods (Komatsu and Hossain 2013). No reports exist about the identification of tissue-specific proteins in millets. More focus is required on identifying tissue-specific proteins in millets, particularly those expressed under abiotic stress conditions in root and growing spike tissues. These proteins aid in identifying the crucial proteins related to climate resilience and nutrient enrichment.

Stress-responsive protein identification

The main obstacles to crop production are abiotic stresses such as salt, drought, flooding, and extreme heat or cold. In response to abiotic stress, plants regulate their proteins, including up-regulating stress-response proteins and synthesizing new ones, to enhance defense mechanisms and promote stress tolerance. There have been reports on the cellular mechanisms behind stress sensing and signal transduction into cellular organelles (Sekine et al. 2021; Ebeed and Ahmed 2023). However, each organ has a different way that plant cells react to abiotic stresses. Several studies were reported on the identification of proteins under stress conditions; these include heat and water stresses in soybeans (Katam et al. 2020), cadmium stress in wheat roots (Jian et al. 2020), cytokinin and abiotic stimuli in barley root (Berka et al. 2020), heat-accumulated proteins in moss plant (Guihur et al. 2021), repeated drought stress in maize (Schulze et al. 2021), osmotic-stress-related proteins in rice (Zang and Komatsu 2007), various abiotic stresses in legumes (Jan et al. 2023). Several other studies were also reported in other plants for the identification of proteins under various stresses, viz., cold, water-lodging,

Table 2 Details of stress-responsive protein profiling studies reported in millets

Name of the millet	Type of stress	Plant tissue/part	Proteomics techniques used	Changes in proteome	Functions of proteins identified under stress	References
Foxtail millet	Drought	Seedlings	LC-MS/MS analysis	Totally 252 up-regulated proteins and 69 down-regulated proteins	Defense responses; photosynthesis, carbon metabolism, ROS scavenging, protein synthesis, etc	Pan et al. (2018)
	Drought	Mature seeds	2D-electrophoresis	Several up-regulated storage proteins and drought stress-responsive proteins	Biochemical and metabolic pathways, including storage protein, amino acid and protein metabolisms, energy metabolism, response to stress and defense, and chaperones	Xu et al. (2020)
	Drought	Seeds	2D-electrophoresis MALDI-TOF/TOF	Totally 57 up-regulated and 47 down-regulated proteins	Storage proteins, protein folding, starch and sucrose metabolism, glycolysis/gluconeogenesis, biosynthesis of amino acids, detoxification and defense, protein degradation, tricarboxylic acid (TCA) cycle, protein synthesis, energy metabolism, transporter, pentose phosphate pathway, and signal transduction	Li et al. (2019)
	Drought	Leaf and root (drought and recovery treatments)	LC-MS/MS analysis	Drought treatment: 773 proteins in leaf and 649 proteins in root Recovery treatment: 952 proteins in leaf and 779 proteins in root	Drought treatment-leaf: metabolic processes; root: maintaining the stability of root cell environments Recovery treatment-leaf: photosynthesis and leaf development; root: regulating secondary metabolite metabolism and root cell development	Gao et al. (2023)
Pearl millet	Salinity	Seedlings	2D-electrophoresis MALDI-TOF/MS	29 differentially expressed, salt-responsive proteins	Signal transduction, photosynthesis, cell wall biogenesis, stress-related and several metabolisms like energy, lipid, nitrogen, carbohydrate and nucleotide metabolisms	Veeranagamallaiah et al. (2008)
	Salinity	Roots (An04-salt susceptible; Yugu2- salt tolerant)	LC-MS/MS analysis	484 proteins-An04 633 proteins-Yugu2 789 phosphoproteins- An04 731 phosphoproteins- Yugu2	Phosphoproteins-signal transduction, regulation of gene expression, translocation, ion transport, and metabolism processes	Pan et al. (2021)
	Drought	Root, leaf and seed	LC-MS/MS	Protein change: leaf tissue-120, roots-25, seeds-10	Cell wall-, lipid-, secondary- and signaling metabolism and the concomitantly observed increased root length point to an impaired shoot-root communication under drought stress	Gao et al. (2023)
	Drought 843-22B-susceptible ICTP8203-tolerant (Comparative proteome with wheat)	Leaf, root and seed		Totally 4564 proteins were identified in pearl millet and 7994 proteins were identified in wheat	Genotype 843-22B-transport and stress-related proteins in the root, mitochondrial electron transport, TCA cycle, C1-metabolism in leaf Genotype ICTP8203-Drought stress enhanced protein functions for cell wall degradation, signaling and polyamine metabolism in the root, light reactions, photorespiration, transport and signaling in leaf and a strong regulation was observed in the development and polyamine metabolism in seed tissue	Ghatak et al. (2021)
Finger millet	Salinity	Seedlings	2D-electrophoresis	Total of 295 and 315 protein spots for the tolerant (Tol) and sensitive (Sen) accessions, respectively	–	Jha (2022)
	Drought	Leaf	LC-MS/MS	Totally 3009 differentially expressed proteins were identified	Hydrolase activity, glycosyl bond formation, oxidoreductase activity, carbohydrate binding and biosynthesis of unsaturated fatty acids	Li et al. (2021)

Details of stress type, plant part or tissue used, type of proteomic tool employed, changes in protein count and functions of expressed proteins are added with references

heat, drought, high temperature, salinity, etc. (reviewed in (Sharma et al. 2019)).

Among millets, many studies were reported on stress-responsive protein identification in the foxtail millet (Table 2). Most studies were reported on the identification of proteins under drought-stress conditions. These include the identification of 252 up-regulated proteins and 69 down-regulated proteins in seedlings (Pan et al. 2018), several up-regulated storage proteins and drought stress-responsive proteins in seeds (Xu et al. 2020), 57 up-regulated and 47 down-regulated proteins in seeds (Li et al. 2019), and 773 proteins in leaf and 649 proteins in root tissues (Gao et al. 2023). Two studies were also reported on protein identification under salinity treatment with 29 differentially expressed, salt-responsive proteins in seedlings (Veeranagamallaiah et al. 2008) and 484 proteins in susceptible genotype (An04) and 633 proteins in tolerant genotype (Yugu2) in roots (Pan et al. 2021). Following foxtail millet, pearl millet has a few reports on protein identification under drought and salt stress (Table 2). Finger millet is the only other millet having report on the stress responsive (drought) protein identification in leaf with a total of 3009 differentially expressed proteins (Li et al. 2021). Other millets have no reports to date on the identification of stress-responsive proteins.

Plant species such as *Arabidopsis thaliana*, *Brassica rapa*, *Brassica oleracea*, and *Zea mays* have enormous data for stress-specific protein identification, and these can be mined from the plant proteome data base (PPDB) (Fig. 1). (<http://ppdb.tc.cornell.edu/dbsearch/searchsample.aspx>). The stress-responsive proteome of no millet has been added to PPDB to date. This also hampers the mining of any stress-responsive proteins in minor millets, although they possess excellent climate resilience features, especially for heat, drought, and high temperatures. High-resolution proteome studies, especially in foxtail millet and finger millet, might help mine key proteins of stress tolerance. Even this study can be conducted in proso millet, which has a reference proteome. This would provide the lead for understanding and improving stress tolerance in other cereals by manipulating the expressions of key proteins using biotechnological tools.

Protein localization studies

Characterizing a protein's function within a cell depends heavily on its subcellular (organelle) location. The functional machinery of the cell, which consists of proteins, must be present in particular cellular compartments for optimal functioning. Protein localization errors can impair processes like metabolic pathways, defense responses, and signaling, leading to stunted growth and, in severe cases, cell death, ultimately contributing to crop yield loss (Sahu et al. 2020). Subcellular localization is therefore a crucial characteristic in the functional characterization of proteins

(Park et al. 2011). The sub-cellular localization of proteins is a highly challenging task that demands high skill, high cost, and sophisticated instruments like confocal microscopes (Fig. 1). This becomes more complex in the case of multi-localized proteins (MLPs), which are found in several subcellular organelles and whose ability to move between places is essential to their cellular function or allows them to perform various tasks in the cell according to the physiological condition (Christoforou et al. 2016). A variety of proteomics tools are being used to identify the location and quantity of proteins in intricate subcellular organelles and structures. These approaches include organelle purification (Ambastha et al. 2021), protein proximity labeling (Jonas and Mason 1981), mass spectrometry-coupled fluorescence imaging (Ambastha et al. 2021), cell-wide biochemical fractionation (MS), and more recently CRISPR based imaging (Christopher et al. 2021).

Imaging techniques provide information about subcellular protein distribution in whole cells. Imaging has been an increasingly effective method for establishing a genetic connection to phenotype, including subcellular location alterations, thanks to recent developments in genome-wide RNA interference technologies and CRISPR-Cas9-based tools (De Groot et al. 2018). There are two primary methods for fluorescence imaging that can be used to obtain subcellular information: immunocytochemistry, which uses fluorescently tagged affinity reagents to bind particular proteins and enable visualization, and live-cell imaging, which involves modifying the protein of interest to express a fluorescent tag for detection (Christopher et al. 2021). Scores of reports are available on the subcellular localization of proteins in the model plant *Arabidopsis thaliana*. As a result, proteome-scale *Arabidopsis* protein subcellular location data has been generated (<https://suba.live/>) (Hooper et al. 2017). Even several localization studies were also reported in major cereals like rice (Krishnan et al. 1992; Martín and Busconi 2000; Yu et al. 2015; Takahashi et al. 2019), maize (Lending et al. 1988; Goday et al. 1994; Bihn et al. 1997; Chikwamba et al. 2003), wheat (Manning and Ciuffetti 2005; Xu et al. 2009; Wei et al. 2021), and other crops. A collection named Crop Proteins with Annotated Locations (cropPAL; <http://cropPAL.org>) was created, which provides information about the subcellular locations of proteins found in barley, wheat, rice, and maize (Hooper et al. 2016). In silico analysis of protein localization refers to the computer prediction or evaluation of the spatial distribution of proteins within a cell. This entails employing bioinformatics tools, algorithms, and databases to infer the likely cellular compartments or organelles in which a protein may be located. In silico analysis, tools such as LOCALIZER (Sperschneider et al. 2017), Plant-mPLOC (<http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/>) (Chou and Shen 2010), and Plant-mSubP (Sahu et al. 2020) were created for predicting the localization of proteins

in plants. In silico analysis of protein localization, various methodologies and strategies are utilized as sequence-based prediction (Ebeed 2022), targeting signals and motifs that direct proteins to particular organelles or compartments (Ebeed et al. 2018), homology-based methods to predict localization by comparing protein sequences to those with known localizations (Sinha et al. 2018), and protein–protein interaction networks that indicate the subcellular localization if a protein interacts with others known to be in a specific compartment (Jiang and Wu 2012). It is important to note that in silico analysis of protein localization has limitations and may not always accurately predict the actual localization of a protein. Experimental validation using techniques such as immunofluorescence, microscopy, or protein fractionation is essential to confirm the predicted localization.

Millet is lagging behind major cereals in protein localization studies. The location of the WD protein was found in foxtail millet, and the SiWD40:GFP fusion protein was found in the nucleus of tobacco mesophyll cells (Mishra et al. 2012). WD40 is a key protein expressed during stress conditions. The researchers investigated the sub-cellular localization of a stress-responsive protein called late embryogenesis abundant (LEA) in foxtail millet. They achieved this by combining LEA14 with GFP and introducing it into onion epidermal cells and foxtail millet root protoplasts. The protein was found to be present in both the nucleus and the cytoplasm (Wang et al. 2014). The subcellular location of 11 pentatricopeptide repeat (PPR) proteins was evaluated

by fusing the PPR genes with GFP and expressing them in protoplasts of foxtail millet (Liu et al. 2016a). Localization of LIM domain proteins was detected by GFP fusion and expression in tobacco protoplasts and they were mainly distributed in both the cytoplasm and nucleus (Yang et al. 2019). The localization of the stress-responsive protein Argonaute 1 (AGO1) in the nucleus was confirmed through. AGO1b-GFP fusion and expression in foxtail millet protoplast (Liu et al. 2016b). Apart from these, not many reports are available on the sub-cellular localization studies in millets with experimental methods. However, a few in silico reports are available on the analysis of the localization of proteins in intracellular organelles in millets. For example, the Wolf-psort (<http://www.genscript.com/wolf-psort.html>) tool was used to predict the subcellular localization of the stress-responsive proteins of foxtail millet (Pan et al. 2018). In finger millet, localization of jasmonate ZIM-domain (JAZ) protein family was analyzed by similar in silico tools (Sen et al. 2016). We have also predicted the localization of phosphate transporter 1–5 (PHT1–5) family proteins of foxtail millet by the Plant-mPLOC tool (Chou and Shen 2010) (<http://www.csbio.sjtu.edu.cn/bioinf/plant/>). The results resemble the reported analysis of PHT1–5 family members in other plants by experimental methods (Fig. 2).

Protein localization studies of millets need more attention to further understand their subcellular location and working mechanism, especially under stress conditions. This will help dissect the roles of specific proteins involved in drought

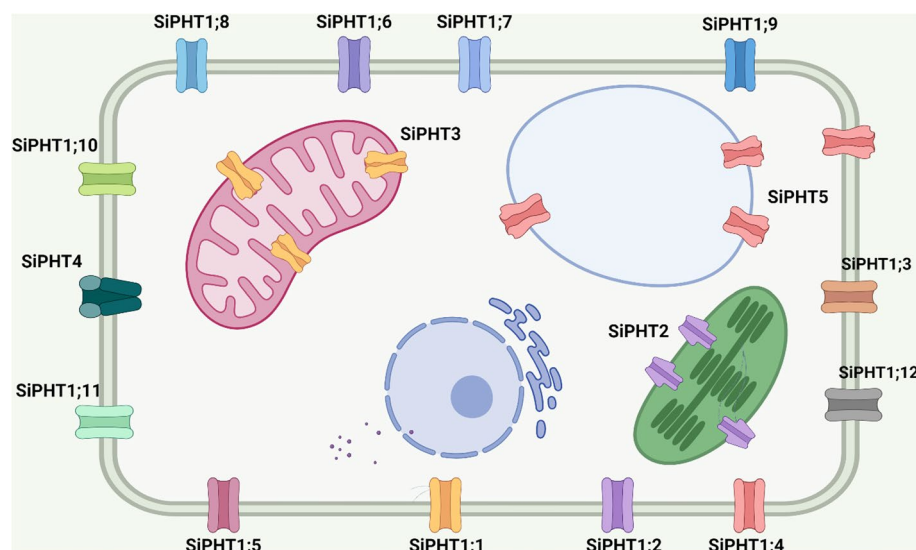


Fig. 2 In silico analysis of localization of phosphate transporter (PHT) 1 to 5 of foxtail millet (*Setaria italica*) (SiPHT1 to SiPHT5) as analyzed by Plant-mPLOC (Chou and Shen 2010) (<http://www.csbio.sjtu.edu.cn/bioinf/plant/>), a web-enabled tool. PHT1, PHT2, PHT3, PHT4, and PHT5 family transporters of foxtail millet are involved in the acquisition and intracellular transport (Ceasar 2020) of phosphorus. The SiPHT1 family has 12 members (SiPHT1; 1 to 1; 12) that

were reported earlier (Ceasar et al. 2014). All 12 members were predicted to be localized to the cell membrane. PHT2 is localized to the chloroplast, PHT3 is localized to the mitochondria, PHT4 is localized to the cell membrane and PHT5 is localized to the vacuolar envelope and cell membrane. The protein sequences used are included in the supplemental file 1

tolerance and nutrient transport. For localization studies, the full coding sequence of the desired protein must be cloned along with reporter genes such as GFP and GUS. These genes make it possible to see where the protein is located. But many millets lack a complete and annotated genome sequence. So, localization studies can be performed by cloning the gene for any protein of interest. But for other millets, it is very difficult to find coding sequences and construct the plasmids. The lack of quick transformation methods and longer generation times for millets are other bottlenecks for designing the localization studies. Protoplast transformation is a quicker method for localizing proteins compared to whole-plant transformation. Even in past reports, millet proteins were located by transforming the protoplasts of other plants, like tobacco. Foxtail millet was the only millet frequently used for protein localization by the protoplast transformation. Many millets need protocols for the transformation of protoplasts, as it might help to detect protein locations via protein-GFP construct delivery. This would lead to the dissection of specific locations and functions of key proteins involved in climate resilience and nutrient supplementation.

Protein structure determination

Determining the crystal structure of a protein is highly significant in multiple scientific disciplines (Varadi and Velankar 2022). The three-dimensional structure of a protein can reveal essential information about its function (Kuhlman and Bradley 2019). Proteins are crucial for plant growth, development, and stress responses. Studying plant protein structures can help modify them to create improved crops with features like stress tolerance or enhanced nutritional value. Protein architectures have implications beyond their direct uses, aiding in the comprehension of molecular biology and essential plant activities such as growth, yields, and stress tolerance. They offer information on the correct configuration, interactions with other molecules, and the significance of these interactions in plant functions. Prediction of the accurate 3D structure of a protein from its amino acid sequence is challenging in biology and methods such as X-ray crystallography or NMR spectroscopy are routinely used to determine the structure of the proteins in laboratories (Fig. 1) (Velankar et al. 2021) (Agnihotry et al. 2022). These methods are very expensive, time-consuming, and often fail to produce results for complex proteins. The structures determined by the wet lab experiments are deposited at the Protein Data Bank (PDB) (<https://www.rcsb.org/>). Since protein structure determination using wet lab methods is highly challenging, many computer-aided tools were developed to predict the 3D structures of the proteins. So structural biologists and computer engineers developed several tools to

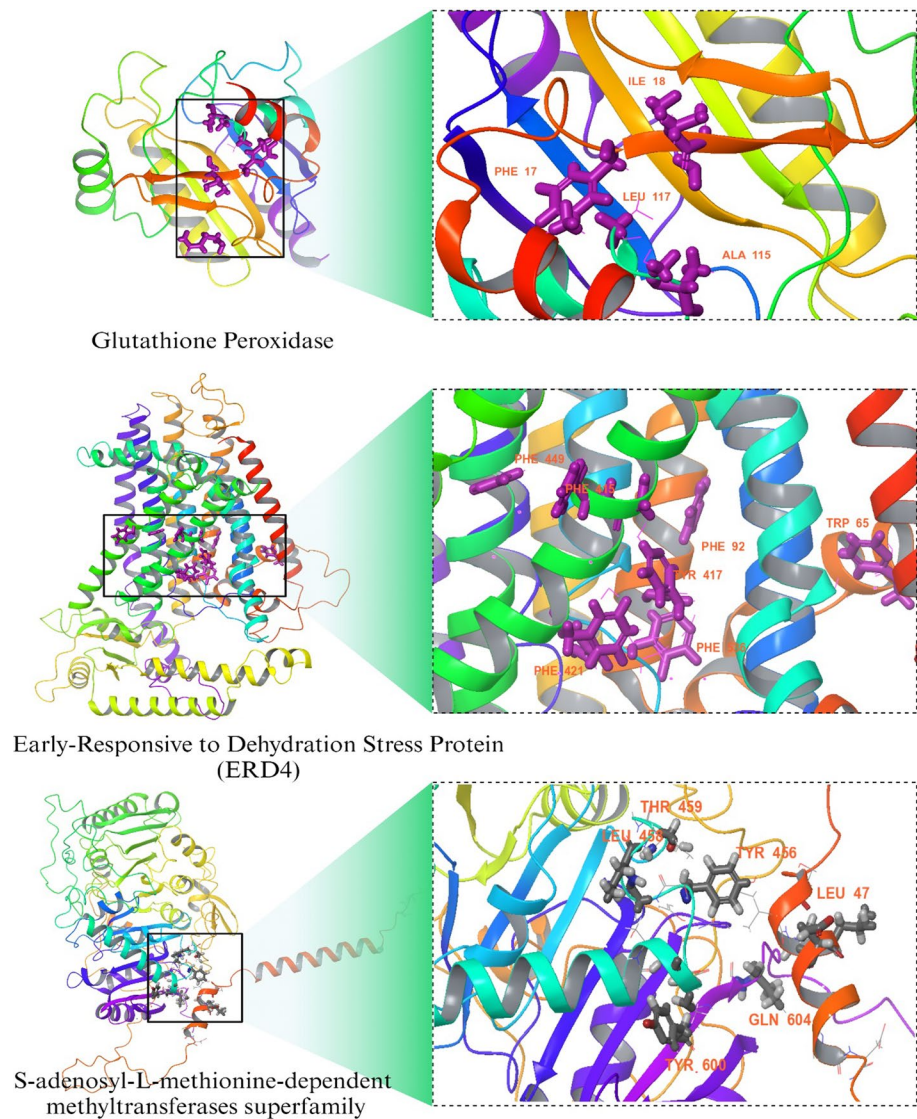
predict the structures from amino acid sequences (Kuhlman and Bradley 2019).

The experimental determination of crystal structures is lagging behind for plants in general and millets in particular, and only a limited number of plant protein structures were determined compared to other groups of organisms (Table 1). Among cereals, rice, wheat, and maize have 100 or more crystal structures experimentally determined and deposited at PDB (<https://www.rcsb.org/>) to date. Among millets, major millet sorghum has the highest number of 40 experimentally determined structures so far. The experimentally determined structures available for sorghum include hydroxynitrile lyase (2.3 Å) (Lauble et al. 2002), dhurrinase 1 (1.8 Å) (Lauble et al. 2002), and hydroxycinnamoyltransferase (2.012 Å) (Walker et al. 2013). Other millets have fewer or no experimentally determined structures, with 7 protein structures for pearl millet, 4 structures for foxtail millet, 3 structures for finger millet, and 1 each for proso and barnyard millets. The pearl millet has experimentally determined structure for the important antioxidant enzyme dehydroascorbate reductase (2.51 Å) (Krishna Das et al. 2016). Similarly, the experimentally determined structure for ferritin (2.20 Å) is available for foxtail millet, which gives insight into understanding plant ferritins (Wang et al. 2023). In finger millet, an experimentally determined proteinase inhibitor structure is available (by NMR spectroscopy) (Wang et al. 2023). The experimentally determined crystal structure of peroxidase (1.79 Å) is available for proso millet (<https://doi.org/10.2210/pdb7dlh/pdb>) and the structure of an antimicrobial peptide (by NMR) is available for barnyard millet (Nolde et al. 2011). There is no experimentally determined crystal structure reported for other minor millets such as kodo millet, browntop millet, and tef.

Apart from the experimental determination of the 3D structures of proteins, many computer-aided tools were also used for plant protein structure prediction. An artificial intelligence (AI)-based protein structure prediction tool, AlphaFold, was introduced for quick and accurate prediction of protein structures (Jumper et al. 2021; Baek et al. 2021). For example, several thousand protein models were predicted by AlphaFold for rice and maize (Table 1). Among millets, sorghum has the highest number of 179 protein models created by AlphaFold, and other millets have only a few models (Table 1).

We have also predicted the protein models of key abiotic stress tolerance proteins reported (Kosová et al. 2018) for foxtail millet by AlphaFold (Fig. 3; Supplemental Fig. S1). The abiotic stress tolerance proteins of foxtail millet, such as glutathione peroxidase (GPx), early-responsive to dehydration stress (ERD4), S-adenosyl-L-methionine-dependent methyltransferases (SAM-Mtases) superfamily, heat shock protein 90 (HSP90), superoxide dismutase Cu–Zn-related (SOD-CuZn) and stress protein, were modelled by

Fig. 3 Homology models of abiotic stress-related proteins of foxtail millet (*Setaria italica*) with the active residues, which are primarily involved in abiotic stress responses. The modeled stress proteins are glutathione peroxidase (AF-A0A4U6V9V6-F1), early-responsive to dehydration stress (ERD4) (AF-K3XQ47) and the s-adenosyl-L-methionine-dependent methyltransferases superfamily (AF-A0A4U6VVU4). All proteins were modeled using AlphaFold (Jumper et al. 2021). The active site residues are projected. The modelled proteins were subsequently visualized in Maestro 13.8 (Schrodinger suite). The protein sequences used are included in the supplemental file 2



AlphaFold. Subsequently, the active site residues of these proteins were identified with the help of a sitemap in the Maestro tool. GPx plays a key role in plants under abiotic stress conditions such as drought, salinity, extreme temperatures, heavy metals, etc. In particular, it contributes to the antioxidant defense system and protects plant cells from oxidative stress by scavenging reactive oxygen species (Hasanuzzaman et al. 2017). ERD4 is involved in a range of cellular activities such as osmotic adjustment, antioxidant defense, stomatal regulation, transcriptional regulation and hormone signaling, allowing plants to survive under water-deficient conditions (Slawinski et al. 2021). SAM-Mtases superfamily plays a key role in several biological processes, such as secondary metabolism, hormone signaling, protein function, and gene expression. S-Adenosyl-L-methionine (SAM) is a key molecule in eukaryotic cells. SAM is involved in different pathways, such as ethylene, lignin

biosynthesis and responses to environmental stresses, and it has essential roles in several cellular processes (Sauter et al. 2013). HSP90 is known as a stress-inducible protein where it promotes stress tolerance, maintains cellular homeostasis and also assists in the proper folding and stabilization of stress-responsive proteins under the conditions of heat shock, cold stress, drought, salinity and oxidative stress. Most importantly, it helps plants adapt and evolve as they grow in new environments (Ul Haq et al. 2019). SOD-CuZn which is present in the cytosol, chloroplasts and peroxisomes, plays an important role in plant stress. SOD-CuZn supports the detoxification process, reducing cell damage caused by heavy metals, extreme temperature and superoxide radicals (Xu et al. 2012). The designed structures were further analyzed to identify the functionally important residues of stress response proteins that actively function in plants under biotic and abiotic stresses. The identified functional residues

identified in these models can be used to trigger the activity and stress response of plants under stressful conditions, leading to the development of a stress mitigation strategy. Consistent with this statement, molecular docking research was carried out on *Oryza sativa* receptors responsible for abiotic stress, using twelve pyrabactin resistance I and seven type 2C protein phosphatases as targets (receptors) and abscisic acid as a ligand (Niranjan et al. 2021). Using recombinant techniques such as CRISPR, stress responses in plants can be modulated by targeting the residues of stress proteins screened in this review. In recent reviews, CRISPR/Cas tools have been projected to help engineer the residues of the proteins involved in nutrient uptake (Ceasar et al. 2022, 2023). Furthermore, the stress tolerance ability of plants can be improved through the use of agrochemicals that can be discovered through protein–ligand docking methods by targeting these residues (Fig. 3 and Supplementary Fig. S1). With increasing accuracy, AI tools like AlphaFold will be useful to predict the structures of a wider range of millet proteins, including minor millets such as kodo millet, little millet, and tef. AI-based tools may be useful for predicting protein dynamics and interactions, providing a more complete picture of protein function in stress responses in millets.

More efforts need to be undertaken by structural biologists to find the experimentally proven structures of the millet proteins, especially those involved in abiotic stress tolerance and nutrient transport. This will help us to understand any key residues and novel mechanisms of operation under stress conditions compared to existing models. Characterizing the structures of nutrient transporters is another key area of structural biology research for millets since they acquire and transfer comparatively higher amounts of nutrients to seeds (especially calcium from finger millet) than other cereals like rice and wheat. Moreover, characterizing the structures and mechanisms of seed-specific transporters would help understand the grain filling mechanism and can help enhance the nutrient fortification in other cereals like rice and wheat.

Conclusion and future prospectus

It is very difficult to meet the food demand of the growing world population, especially in an ever-changing climate. Apart from improving the climate resilience of existing stable crops, it is also necessary to study and adopt less-researched crops like millets. Millets have been given much importance in recent years due to their excellent climate resilience and nutrient enrichment features. The omics (genomic, transcriptomic, metabolomic and proteomic) studies of millets are lagging behind those of major cereals. Especially, only a little effort has been made on proteomics studies in millets, although they possess excellent climate

resilience properties. Modern proteomic approaches may readily uncover the molecular basis of stress responses and assess the functioning of translated genomic regions. Identifying and verifying stress-responsive proteins is a viable technique for breeding better crops in the future. Several model plants have numerous protein resources and have reference proteome data. There are two reference proteome data sets available for foxtail and one each for sorghum and proso millet. Nevertheless, reference proteome data is not available for millets such as pearl millet, finger millet, barnyard millet, kodo millet, tef, and browntop millet. Sorghum, foxtail millet, and little millet, each have several thousand protein sequence entries at UniProt. Other millets have a few hundred protein listings. In particular, kodo millet has no protein entry in UniProt. Although genes are the storehouse of information for protein synthesis, proteins are the workhorse of the cell, executing several functions. Since many proteins undergo dynamic changes after translation and even delivery, it is necessary to confirm the activity of particular proteins in particular tissues. No reports exist about the identification of tissue-specific proteins in millets. More focus is required on identifying tissue-specific proteins in millets, particularly those expressed under abiotic stress conditions in tissues like roots for abiotic stress tolerance and growing spikes for nutrient filling. Some reports are available on the identification of stress-responsive proteins in foxtail millet. Most studies focused on identifying proteins under drought stress. Pearl millet has had protein identification studies conducted in drought and salinity environments. Millets are not as advanced as the main cereals in protein localization research. Some studies have focused on sub-cellular localization of proteins in foxtail millet, but there is a lack of data on sub-cellular localization investigations of proteins in other millets. To date, rice, wheat, and maize have 100 or more experimentally determined and deposited crystal structures at PDB (<https://www.rcsb.org/>). Currently, sorghum possesses the highest number of experimentally determined crystal structures, with a total of 40. Pearl millet has 7 protein structures, while foxtail millet has 4, finger millet has 3, and proso, little millet and barnyard millets have 1 structure each. Kodo millet and tef lack any submitted crystal structure.

Future protein research in millets holds significant importance due to several reasons. Understanding the protein profile of millets can provide insights into their nutritional composition, including amino acid content and protein quality. Millets are hardy crops that can grow in diverse agro-climatic conditions, including regions with low rainfall and poor soil fertility. Research in millet proteomics can help in developing varieties with improved nutritional content, yield, and resilience to biotic and abiotic stresses. This contributes to food security, especially in regions prone to climate change and food insecurity. Millets are attracting

global attention because of their potential health advantages, such as being gluten-free, having a low glycemic index, and being rich in dietary fiber, vitamins, and minerals. Protein research can identify specific proteins associated with these health-promoting properties, aiding in the development of functional foods and nutraceuticals. Protein research can help identify allergenic proteins in millets, facilitating the development of hypoallergenic varieties. Additionally, it can contribute to ensuring the safety of millet-based products by detecting any harmful proteins or contaminants. Understanding the protein profile of millets can aid in crop improvement programs. By identifying key proteins associated with desirable traits such as drought tolerance, pest resistance, and nutrient uptake, researchers can accelerate the breeding process to develop improved millet varieties. Protein research can also provide insights into the biochemical pathways involved in millet grain development, germination, and post-harvest processing. This knowledge can guide efforts to optimize processing techniques, enhance product quality, and minimize nutrient loss during storage and cooking. Millets are known for their low water and input requirements compared to major cereal crops like rice and wheat. By understanding the protein basis of millet's resilience and adaptability, researchers can promote the cultivation of millets as part of sustainable agricultural practices, contributing to water conservation and environmental sustainability. Overall, research on millet proteins is essential for maximizing the nutritional, agronomic, and economic potential of these versatile grains, thereby addressing global challenges related to food security, public health, and sustainable agriculture.

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Data availability All data generated or analysed during this study are included in this published article.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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